

# Transformer-Based Universal Source Coding

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Active research notebook. Things are still getting cleaned up and results are still coming in. Documenting the toolkit and experimental pipeline as it gets built.

The central question: can a small transformer trained offline on IID binary sequences produce better probability estimates than a Laplace estimator, particularly at small sequence lengths where Laplace is known to be slow to converge?

## 1 Motivation

Universal source coding is the problem of compressing a sequence from an unknown source. The classical approach is to estimate symbol probabilities sequentially from the data and encode using arithmetic coding. The Laplace estimator does this as follows: after observing  $k$  ones in  $n$  symbols, it assigns  $\hat{p}_1 = \frac{k+1}{n+2}$ .

The limitation is that at small  $n$ , this estimate is noisy and the BPS overhead above the entropy rate is non-trivial. The theoretical overhead for sequential universal source coding scales as  $\frac{1}{2} \frac{\log N}{N}$  bits per symbol, fine asymptotically, not great at  $N = 10$ .

The hypothesis is that a transformer trained on IID binary sequences with uniformly sampled  $p$  has implicitly learned the Bayes-optimal predictor over this mixture. At inference time it can amortize its “warm-up” cost across training sequences and produce tighter probability estimates from the first few symbols, even on IID data with no temporal structure to exploit.

## 2 Information Theory Primitives

Before getting into the compression experiments, the toolkit builds up the core information-theoretic functions from scratch. These serve as both a sanity check and as the measurement infrastructure for the compression benchmarks.

```
import numpy as np
import matplotlib.pyplot as plt
from math import comb
from itertools import combinations
```

```

def convert_sample_to_pmf(sample):
    values, counts = np.unique(sample, return_counts=True)
    probs = counts / counts.sum()
    return dict(zip(values, probs))

def clean_pmf(pmf):
    return {k: v for k, v in pmf.items() if v > 0.0}

def generate_bin_pmf():
    p = np.random.rand()
    return {0: p, 1: 1 - p}

def iid_generate(p_dict, N):
    return np.random.choice([0, 1], size=N, p=[p_dict[0], p_dict[1]])

def entropy_calc(pmf):
    pmf = clean_pmf(pmf)
    pmf_array = np.asarray(list(pmf.values()))
    return -np.sum(pmf_array * np.log2(pmf_array))

```

## 2.1 KL Divergence

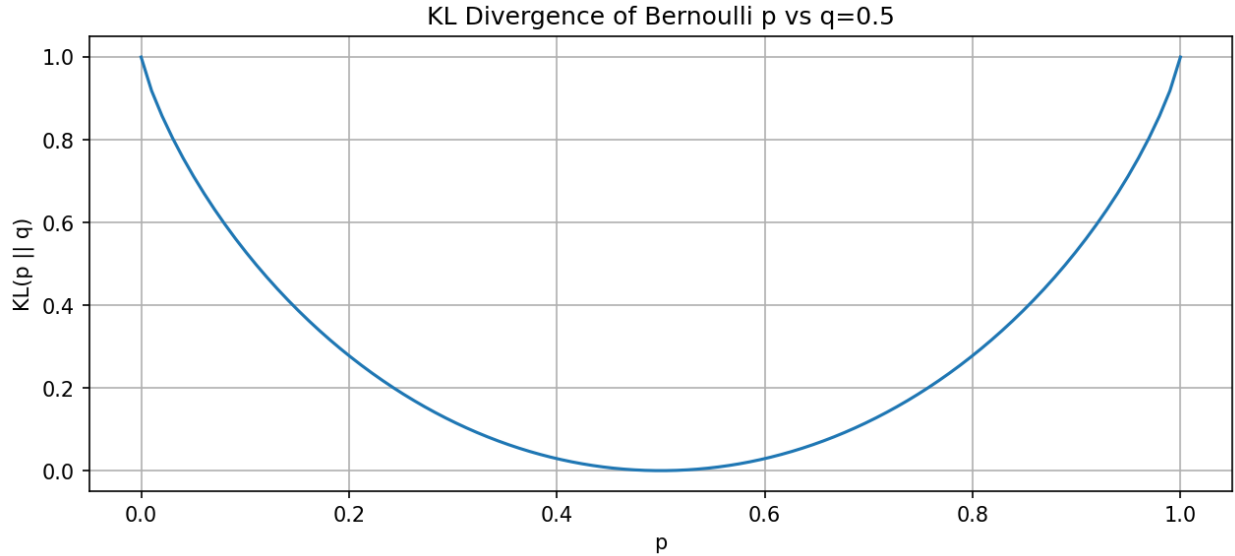
KL divergence  $D_{KL}(p\|q) = \sum_x p(x) \log \frac{p(x)}{q(x)}$  measures how many extra bits are spent encoding a source with true distribution  $p$  using a code designed for  $q$ . It shows up directly in the compression overhead of the Laplace estimator, the BPS excess above entropy is  $D_{KL}(\hat{p}\|p)$  evaluated at the estimated distribution.

The plot below shows  $D_{KL}(\text{Bern}(p)\|\text{Bern}(0.5))$  as a function of  $p$ . It is zero at  $p = 0.5$  (the uniform prior matches the source exactly) and blows up near the extremes.

```

# KL divergence of Bernoulli(p) vs Bernoulli(0.5)

```



### 3 Sequential Universal Source Coding

The core compression metric is bits-per-symbol (BPS) under arithmetic coding. Given a sequence  $x_1, \dots, x_N$  and a corresponding probability array  $p_1, \dots, p_N$  (where  $p_i$  is the estimated probability that  $x_i = 1$  before observing it), the code length is:

$$\text{BPS} = -\frac{1}{N} \sum_{i=1}^N \log_2 p(x_i)$$

For the Laplace estimator,  $p_i$  is updated after each observed symbol. For the transformer,  $p_i$  comes from the model's output softmax given the preceding context. Both schemes are evaluated on the same sequence, the entropy rate  $H(p)$  is the lower bound for both.

```
def sequential_universal_source_coding(sequence, p_array=None):
    """
    Compute expected BPS under arithmetic coding.
    If p_array is None, use the Laplace estimator sequentially.
    """
    n0, n1 = 0, 0
    total_bits = 0.0

    for i, x in enumerate(sequence):
        if p_array is not None:
            p1 = p_array[i]
        else:
            # Laplace estimate
            p1 = (n1 + 1) / (n0 + n1 + 2)

        p1 = np.clip(p1, 1e-10, 1 - 1e-10)
```

```

    p0 = 1 - p1

    if x == 1:
        total_bits += -np.log2(p1)
        n1 += 1
    else:
        total_bits += -np.log2(p0)
        n0 += 1

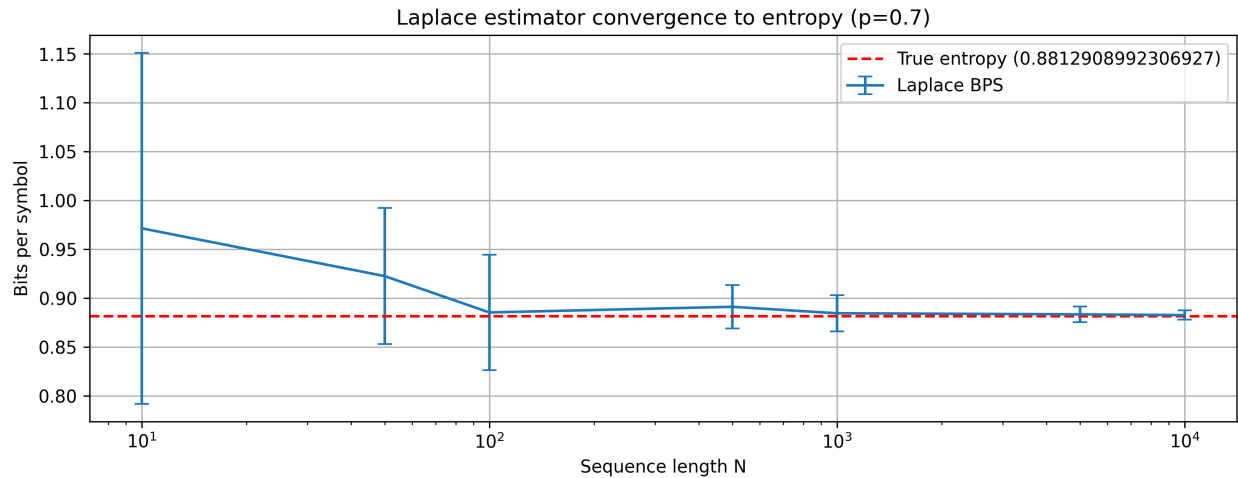
    return total_bits / len(sequence)

```

## 4 Baseline: Laplace Estimator Convergence

Before introducing the transformer, we characterize how the Laplace estimator converges across sequence lengths at  $p = 0.7$ . 100 trials per  $N$  value.

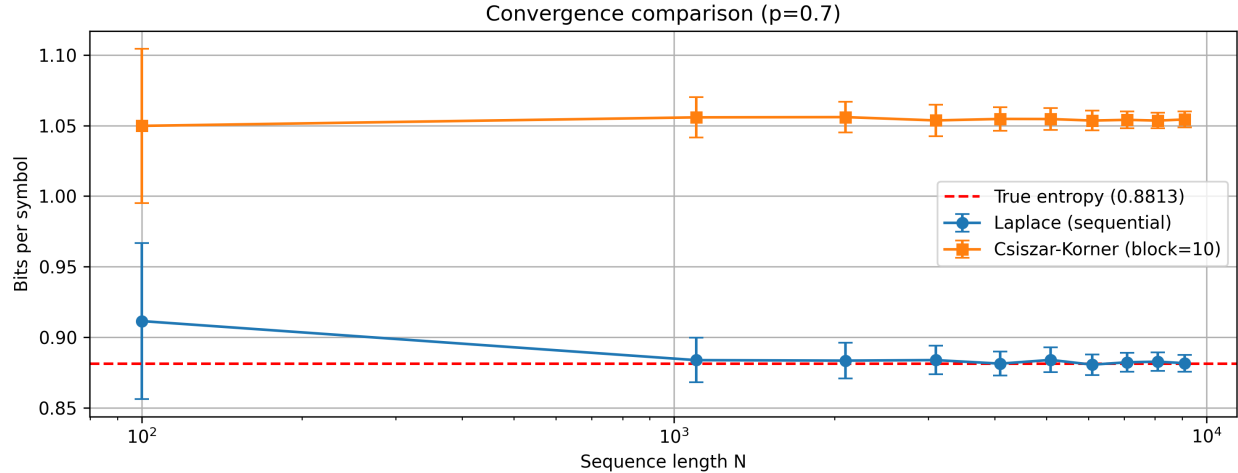
```
# Laplace estimator BPS convergence (p=0.7)
```



The Laplace estimator converges reliably but the variance at  $N = 10$  is large. The mean is already pretty close by  $N = 100$  but the confidence interval still spans roughly 0.1 bits.

We also compare against the Csiszar-Korner block estimator as a secondary baseline. The CK estimator processes the sequence in non-overlapping blocks and estimates  $p$  within each block. With block size 10, it plateaus significantly above entropy even at large  $N$ , the block boundaries introduce a hard floor on performance.

```
# Laplace vs Csiszar-Korner convergence comparison (p=0.7)
```



The CK estimator is clearly worse here, it never converges to entropy in this regime. Laplace wins and is the baseline the transformer needs to beat.

## 5 The Model: minGPT

The transformer used here is minGPT (Karpathy), the **gpt-nano** configuration: 3 layers, 3 attention heads, embedding dimension 48. About 90K parameters total.

Training data: IID binary sequences of length 100, where each sequence is drawn from  $\text{Bern}(p)$  with  $p \sim \text{Uniform}(0, 1)$  sampled fresh per sequence. Training task is standard next-token prediction.

The idea is that at inference time, given the prefix of a new sequence, the model implicitly estimates  $p$  from context and outputs a calibrated next-symbol probability. This gets fed directly into the arithmetic coder in place of the Laplace estimate.

```
import torch
import torch.nn.functional as F
from torch.utils.data import Dataset
from mingpt.model import GPT
from mingpt.trainer import Trainer

class IID(Dataset):
    """IID binary sequences with uniformly sampled p."""
    def __init__(self, n=10, num_samples=10000):
        self.n = n
        self.num_samples = num_samples

    def _sample(self):
        p = torch.rand(1).item()
        seq = torch.bernoulli(torch.full((self.n,), p)).long()
        return seq[:-1].clone(), seq[1:].clone()

    def get_block_size(self):
```

```

        return self.n - 1

    def __getitem__(self, idx):
        return self._sample()

    def __len__(self):
        return self.num_samples

dataset = IID(n=100, num_samples=10000)

model_config = GPT.get_default_config()
model_config.model_type = 'gpt-nano'
model_config.vocab_size = 2
model_config.block_size = dataset.get_block_size()
model = GPT(model_config)

# Training, already run, weights saved
# train_config = Trainer.get_default_config()
# train_config.learning_rate = 1e-4
# train_config.max_iters = 100000
# train_config.num_workers = 0
# trainer = Trainer(train_config, model, dataset)
# trainer.run()
# torch.save(model.state_dict(), 'iid_model.pt')

model.load_state_dict(torch.load('iid_model.pt'))
model.eval()

```

## 6 Inference: Windowed Probability Arrays

For a sequence of length  $N$ , we need  $P(x_i = 1 \mid x_1, \dots, x_{i-1})$  at each position. We use a sliding window: at position  $i$ , feed the model the last `block_size` tokens and take the output logit at the final position. First bit gets a flat 0.5 prior.

```

def transformer_p_array_windowed_fast(sequence, model, device='cpu'):
    model.eval()
    B = model.block_size
    T = len(sequence)
    p_array = [0.5] # uniform prior for first bit, no context

    seq_tensor = torch.tensor(sequence, dtype=torch.long, device=device)

    with torch.no_grad():
        for i in range(1, T):
            start = max(0, i - B)
            context = seq_tensor[start:i].unsqueeze(0)

```

```

logits, _ = model(context)
next_token_logits = logits[0, -1, :]
probs = F.softmax(next_token_logits, dim=-1)
p_array.append(probs[1].item())

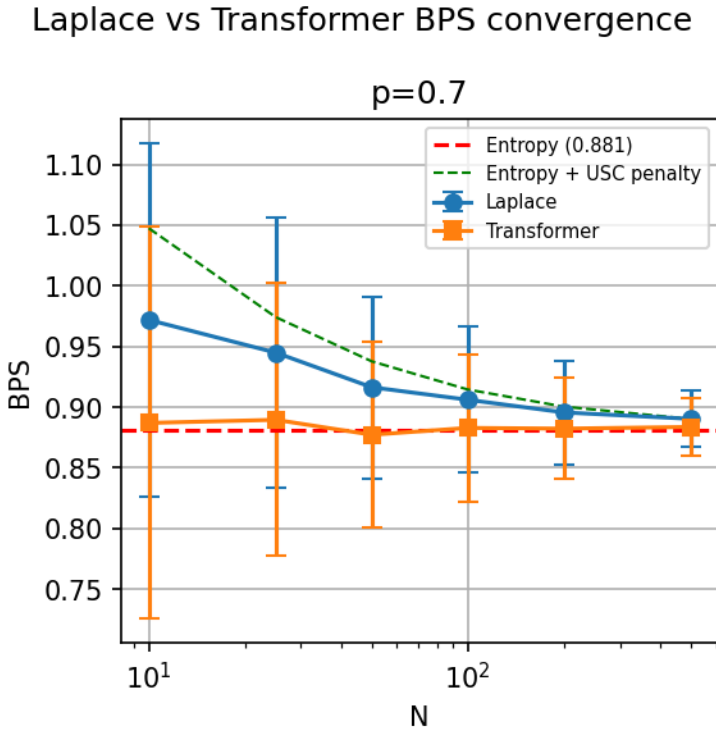
return p_array

```

## 7 Results: $p=0.7$ Pilot

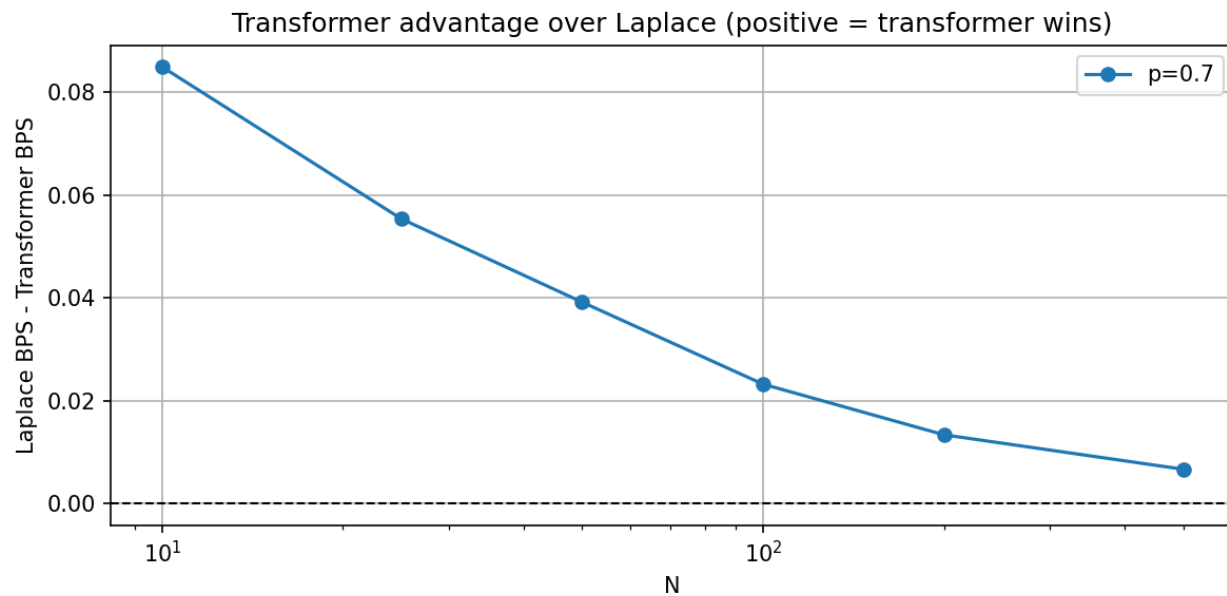
Initial experiment at a single  $p = 0.7$ , sweeping  $N \in \{10, 25, 50, 100, 200, 500\}$  with 100 trials each. The transformer is loaded from the saved checkpoint and run on CPU for reproducibility.

```
# Laplace vs Transformer BPS convergence (p=0.7)
```



The transformer tracks at or below Laplace across all  $N$  values tested. The error bars are large at small  $N$ , this is just variance in the sequences themselves. The gap plot shows the signal more cleanly.

```
# Transformer advantage over Laplace (p=0.7)
```

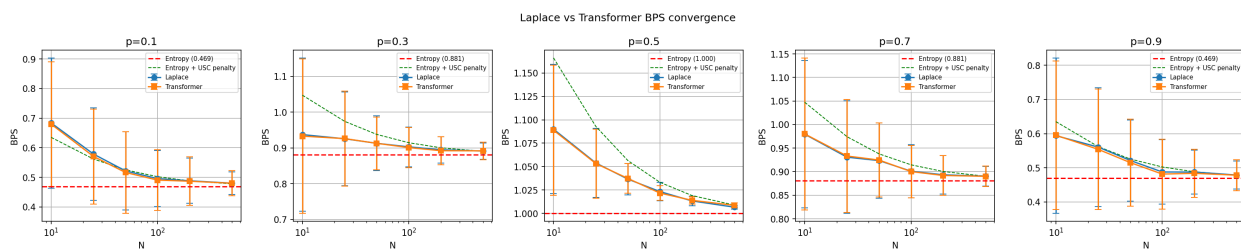


The transformer advantage is strongest at  $N = 10$  ( $\sim 0.085$  bits) and decreases monotonically as  $N$  grows, converging toward zero as Laplace catches up. This is exactly the expected pattern, the transformer's learned prior matters most when there is almost no data to estimate from.

## 8 Results: Fixed $p$ sweep

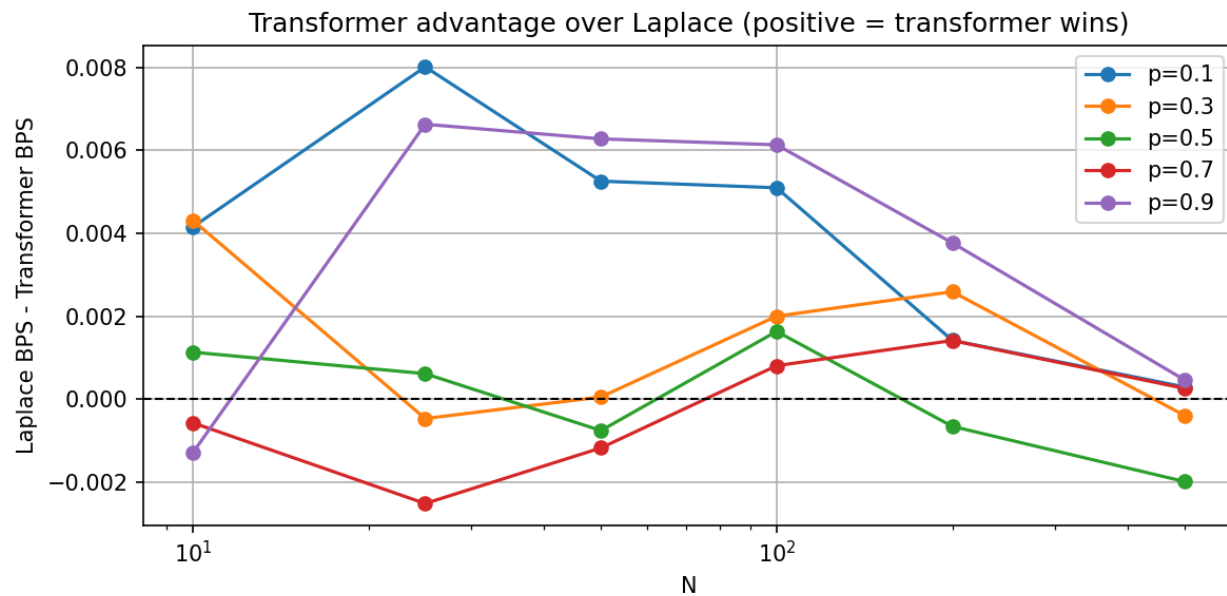
Running the same experiment across  $p \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$ . The  $p = 0.7$  result above used sequences with random  $p$  during training, this run uses fixed  $p$  per trial.

# Laplace vs Transformer BPS convergence (all  $p$  values)



# Transformer advantage over Laplace (all  $p$  values)





The full sweep tells a more complicated story than the  $p = 0.7$  pilot. The gap is noisy and in some cases negative, the transformer occasionally does slightly worse than Laplace, particularly at  $p = 0.5$  and  $p = 0.7$  at larger  $N$ . The  $p = 0.1$  and  $p = 0.9$  cases show the clearest positive gap.

A few things to unpack here:

The symmetric  $p = 0.5$  case is the hardest for the transformer because there is no  $p$  to infer, every sequence looks the same. The Laplace estimator's uniform 0.5 prior is already optimal.

The  $p = 0.1$  and  $p = 0.9$  cases show the strongest advantage because skewed sequences give the model a fast signal about  $p$  even in the first few symbols, while Laplace is still mixing in a symmetric prior.

The negative gaps are small enough (around 0.002 bits) that they are likely within noise, but this needs more trials to confirm.

## 9 Next steps

- Increase the trial count and sequence length
- Train separate models using different probability clusters
- Scale up model size
- make big bucks